Spectral Early Warning Signals for Model Collapse in Recursive Synthetic Training

Anonymous ACL submission

Abstract

During the early stages of recursive synthetic training, spectral indicators of the embedding space suggest the model is improving—even as downstream performance is already degrading. We reveal this misleading signal by tracking spectral properties across 10 recursive generations of Pythia-410M. The result is a reproducible three-phase pattern consistent across three seeds: spectral decompression (generations 0–2), where SLV increases while HellaSwag drops 3.3 percentage points; collapse onset (generations 2–4), where spectral indicators reverse; and accelerating collapse (generations 4–10). Of eight indicators, SLV is the most effective: it requires a single forward pass (< 1 second on one GPU), captures changes across all spectral directions, and declines monotonically after the decompression peak. Varying the real-data mixing ratio across five levels yields a phase diagram localizing $r_{\text{crit}} \in (0.25, 0.3]$, separating monotonic expansion from a transitional regime with initial contraction then recovery. A proportional controller reveals inherent signal attenuation: moderate mixing prevents the spectral contraction that would trigger the controller, rendering reactive feedback ineffective when prevention succeeds. Setting the mixing ratio above the calibrated r_{crit} prevents collapse without real-time monitoring, reducing the problem to one-time offline calibration. The three-phase trajectory reproduces on GPT-2-medium (355M), providing preliminary cross-architecture evidence.

1 Introduction

Synthetic data has become integral to language model training. Instruction tuning, knowledge distillation, and data augmentation all routinely use model-generated text, and as synthetic corpora are recycled into training pipelines, the process often becomes recursive: each generation of models is fine-tuned on outputs from the previous one. Recent work demonstrates that such recursive synthetic training triggers *model collapse*, a progressive erosion of distributional tails and downstream capability (Shumailov et al., 2024; Dohmatob et al., 2024b; Alemohammad et al., 2024). The problem is especially pressing as web-scraped corpora increasingly contain model-generated content, a trend that introduces implicit recursive loops even without deliberate data reuse.

While model collapse is now well-characterized through theoretical analysis (Dohmatob et al., 2024b) and empirical studies (Shumailov et al., 2024; Alemohammad et al., 2024), its effect on the geometric structure of learned representations remains poorly understood. Characterizing how the embedding space contracts during recursive training could yield computationally cheap collapse indicators from internal model states alone, without text generation or benchmark evaluation.

Current collapse detection depends on downstream metrics such as perplexity and benchmark accuracy, or on analysis of generated text through lexical diversity measures. Both approaches require either costly inference or external evaluation infrastructure at every generation. SIGMA (Gu et al., 2026) introduced spectral monitoring through embedding covariance analysis and showed that spectral indicators can track collapse, but validated only on OPT-125M over short generation horizons. Four gaps remain open. First, no work has systematically characterized spectral dynamics beyond a few generations. Second, the temporal relationship between spectral indicators and downstream degradation has not been examined. Third, no study has connected spectral indicators to closed-loop collapse prevention through the lens of early warning signals (EWS) (Scheffer et al., 2009). Fourth, the relationship between real-data mixing ratios and spectral collapse has not been mapped: it is unknown how much real data suffices to prevent spectral contraction, or whether a sharp phase boundary exists.

We address these gaps by tracking spectral indicators of the embedding matrix across 10 recursive generations of Pythia-410M-deduped (Biderman et al., 2023), with pilot validation on Pythia-1B-deduped and cross-architecture confirmation on GPT-2-medium (Radford et al., 2019). Our analysis reveals a reproducible three-phase pattern (Figure 1): spectral decompression (Gen 0–2), where effective rank (r_{eff}) and spectral log-volume (SLV) rise while downstream performance already declines; collapse onset (Gen 2–4), where spectral indicators reverse; and accelerating collapse (Gen 4–10), where contraction accelerates. Among these indicators, SLV is the most effective collapse indicator: it requires only a forward pass on held-out data (no text generation), captures distributed volume changes across all spectral directions, and declines monotonically after its peak. We further design a proportional controller as a testbed for reactive collapse prevention, and find that it exposes an inherent signal attenuation: the rate-based spectral signal degrades continuously as the real-data fraction increases and vanishes entirely above r_{crit}, rendering any reactive spectral controller ineffective in the regime where prevention succeeds.

Our contributions are:

- **C1:** The first multi-seed spectral-downstream characterization over 10 recursive generations, revealing dynamics invisible in prior short-horizon studies and establishing a reproducible three-phase collapse pattern (decompression, onset, acceleration) across 10 generations (§3.2, Table 1).
- **C2:** SLV as a computationally cheap collapse indicator with monotonic post-peak decline and a HellaSwag-spectral asymmetry: downstream degradation begins during the spectral decompression phase, demonstrating that spectral expansion does not indicate model health (§3.2).
- **C3:** Quantitative characterization of signal attenuation under mixing: the spectral controller signal degrades continuously as the real-data fraction approaches r_{crit} and vanishes entirely above it, a limitation robust to threshold reduction (§2.2, §3.3).

Supporting these findings, we construct a spectral collapse phase diagram across multiple mixing ratios that localizes r_{crit} empirically, with the bound robust to both the EMA smoothing parameter and random seed (§2.3, §3.3, §3.5), and provide preliminary cross-architecture evidence that the three-phase trajectory generalizes to GPT-2-medium (§3.6).

2 Method

2.1 Spectral Metrics

Given a language model at generation t, we extract last-layer embeddings for n=1,000 held-out samples to form $\mathbf{E} \in \mathbb{R}^{n \times d}$. Let $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{\min(n,d)}$ denote the singular values of $\mathbf{E}$. We track three complementary spectral indicators:

Spectral Log-Volume (SLV). SLV $= \sum_{i:\,\sigma_i > \epsilon} \log \sigma_i$, where $\epsilon = 10^{-10}$ excludes numerically zero components. When $n < d$ (as in our setting with n=1,000, d=1,024), $\mathbf{E}$ has at most n non-zero singular values; SLV thus measures the log-volume of the occupied subspace rather than the full d-dimensional space. Because it sums over all non-negligible singular values, SLV is sensitive to contraction along any spectral direction, not only the dominant ones.

Effective Rank. $r_{\text{eff}} = \exp(H(\mathbf{p}))$, where $p_i = \sigma_i / \sum_j \sigma_j$ and H is the Shannon entropy, measuring how many dimensions carry substantial variance (uniform $\to \min(n, d)$, concentrated $\to 1$).

Leading Singular Value. σ_1 tracks the dominant spectral direction; rising σ_1 under falling r_{eff} signals representational collapse into a low-dimensional subspace.

2.2 Adaptive Mixing Controller

To test whether spectral rate-of-change signals can drive automated collapse prevention, we design a closed-loop controller that adjusts the real-data mixing ratio r_t at each generation. The update rule is:

$$r_{t+1} = \text{clip}\big(\tfrac{B_{\text{rem}}}{T-t} + k \cdot \max(0,\, \Delta_t - \tau),\; r_{\min},\; r_{\max}\big) \quad (1)$$

where B_{rem} is the remaining real-data budget (fraction of total budget not yet consumed), T is the total number of generations, τ=0.002 is a sensitivity threshold, and k=10 is the controller gain. The signal Δ_t is the baseline-normalized SLV drop: $\Delta_t = (\widetilde{S}_{t-1} - \widetilde{S}_t)/|\widetilde{S}_0|$, where $\widetilde{S}_t$ is the exponential moving average (EMA)-smoothed SLV (α=0.5) and $\widetilde{S}_0$ is initialized from the pretrained model (Gen 0, SLV=2366.68). The signed form

ensures that spectral decompression ($\Delta_t < 0$) does not trigger the controller; only contraction activates the $\max(0, \cdot)$ gate in Eq. 1. The output is clipped to $[r_{\min}, r_{\max}] = [0.20, 0.80]$ and subject to $\sum_t r_t \leq B_{\text{total}}$. The lower bound ensures a minimum synthetic fraction; $r_{\max}$ defines the controller's operational range.

The first term distributes the remaining budget uniformly across future generations; the model degrades gracefully even without spectral signal. The second term injects additional real data when SLV contraction exceeds the threshold τ, proportional to the severity of the spectral drop. This design requires no lookahead and operates with a single hyperparameter choice (k) beyond the budget constraint. We set τ=0.002 as the smallest threshold that avoids false activation during the decompression transient, and k=10 so that a moderate signal ($\Delta_t \approx 2\tau$) adds approximately 2 percentage points of additional real data.

Controller Silence Condition. Define r_{crit} as the minimum constant mixing ratio such that, within T=10 generations, the EMA-smoothed SLV is non-decreasing from the second generation onward: $\widetilde{S}_t \geq \widetilde{S}_{t-1}$ for all $t \in \{2, \ldots, T\}$ (excluding the initial fine-tuning transient at t=1). For conditions with multiple seeds, the condition is evaluated on the cross-seed mean trajectory. When a constant ratio $r \geq r_{\text{crit}}$ is applied, $\Delta_t \leq 0 < \tau$ for all $t \geq 2$ and any $\tau > 0$, so the controller boost is identically zero. This τ-independence means the signal attenuation cannot be resolved by lowering the sensitivity threshold; the same argument extends to cumulative variants. Our experiments (§3.3) empirically bound $0.25 < r_{\text{crit}} \leq 0.3$ over Gen 2–10 for Pythia-410M on OpenWebText.

Second-Order Extension. A natural extension monitors the rate of change of the controller signal itself:

$$\Delta_t^2 = \widetilde{\Delta}_{t-1} - \widetilde{\Delta}_t \tag{2}$$

where $\widetilde{\Delta}_t$ is the EMA-smoothed first-order signal (α_2=0.3). A sustained negative Δ_t^2 indicates decelerating SLV growth, a potential precursor to stagnation even when SLV is still rising. We evaluate this indicator in §3.5.

2.3 Collapse Phase Diagram

We define the *collapse phase diagram* as the mapping from constant mixing ratio r to the SLV trajectory over T generations. For each r, we classify the outcome: *collapse* if SLV declines monotonically after the decompression phase (Gen 2 onward), *stable* if SLV is non-decreasing, and *transitional* if SLV shows non-monotonic behavior after Gen 2 (both increases and decreases within the observation window). The critical mixing ratio r_{crit} (§2.2) is the boundary between collapse and stability in this diagram. We estimate r_{crit} by evaluating multiple constant ratios and identifying the lowest r for which $\widetilde{S}_t \geq \widetilde{S}_{t-1}$ holds for all $t \in \{2, \ldots, T\}$.

Gen	r_{eff}	SLV	PPL	HeSw	D-4	σ_1
0	417.55	2366.68	9.55	.413	.738	1942.8
2	**496.17**	**2467.98**	13.56	.380	.505	718.7
5	479.77	2365.99	24.50	.341	.412	749.2
7	464.10	2293.24	37.82	.319	.375	788.4
10	438.05	2181.35	70.99	.296	.324	874.6

Table 1: Spectral and downstream metrics at selected generations (Pythia-410M, mean of three seeds 42, 43, 44; sample s.d. at Gen 10: r_{eff} ±4.8, SLV ±28.3, PPL ±5.6, HeSw ±.006, D-4 ±.15). Bold marks the spectral peak.

3 Experiments

3.1 Experimental Setup

We use the publicly released final Pythia-410M-deduped checkpoint (Biderman et al., 2023) as our base model and fine-tune on a 50K-subset of OpenWebText. Starting from the pretrained checkpoint (generation 0), each subsequent generation fine-tunes on synthetic text produced by the previous generation, for a total of 10 recursive generations. At each generation we compute the spectral indicators from §2.1 on 1K held-out last-layer embeddings, paired with three downstream metrics: perplexity, HellaSwag accuracy (0-shot), and distinct-4. These three metrics (language modeling capability, natural language understanding, and generation diversity) cover the primary evaluation dimensions accessible to a 410M-parameter model.

For the mixing comparison (§3.3), we evaluate three strategies: (1) **pure collapse** (r_t=0), (2) **fixed mix** (r_t=0.50), and (3) **adaptive** (k=10, Eq. 1). To construct the collapse phase diagram (§3.5), we additionally evaluate fixed mixing at $r \in \{0.2, 0.25, 0.3\}$.

3.2 Spectral Collapse Characterization

Table 1 reports spectral and downstream metrics across 10 recursive generations, which reveal a three-phase structure.

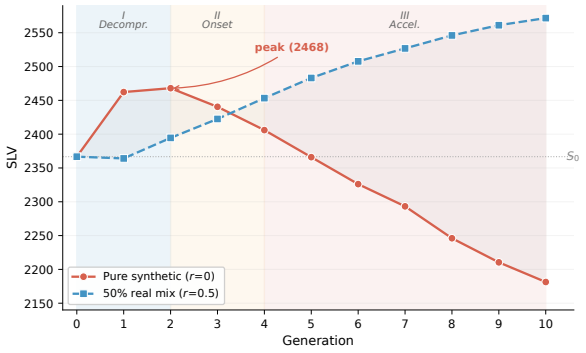


Figure 1: SLV across 10 recursive generations of Pythia-410M (solid: r=0, 3-seed mean; dashed: r=0.5, 2-seed mean) under pure-synthetic training and 50% real-data mixing. Three phases are visible in the pure-synthetic trajectory: decompression (Gen 0–2, SLV peaks at 2468), onset (Gen 2–4), and acceleration (Gen 4–10, SLV falls 11.6% from peak to 2181). Under 50% mixing, SLV rises monotonically (except a negligible Gen 1 dip) to 2572 at Gen 10, 8.7% above baseline and 104 points above the pure-collapse peak.

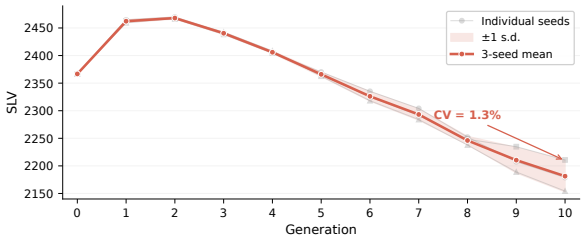


Figure 2: Cross-seed reproducibility of SLV under pure-synthetic training (r=0). Thin lines: individual seeds (42, 43, 44); thick line: 3-seed mean; shaded band: ±1 s.d. The coefficient of variation remains below 1.3% at all generations, confirming that spectral collapse follows a deterministic trajectory.

Phase I: Spectral Decompression (Gen 0–2). r_{eff} rises from 417.55 to 496.2 (+18.8%) and SLV from 2366.7 to 2468.0 (+4.3%), while σ_1 drops from 1942.8 to 718.7. The peak at Gen 2 is consistent across all three seeds (Figure 2; SLV CV $<$ 1.3% at all generations; full per-generation statistics in Appendix Table 5). Crucially, this decompression does not indicate improved model quality: HellaSwag drops 8.0% and distinct-4 falls 31.6% during the same interval (Figure 3), and SLV computation takes $<$1 s vs. ∼70 min for benchmarks.

Phase II: Collapse Onset (Gen 2–4). Both r_{eff} and SLV reverse after Gen 2 (r_{eff}: 496.2→493.6, SLV: 2468→2440), marking the transition from decompression to collapse while perplexity continues its monotonic rise. SLV's post-peak decline is

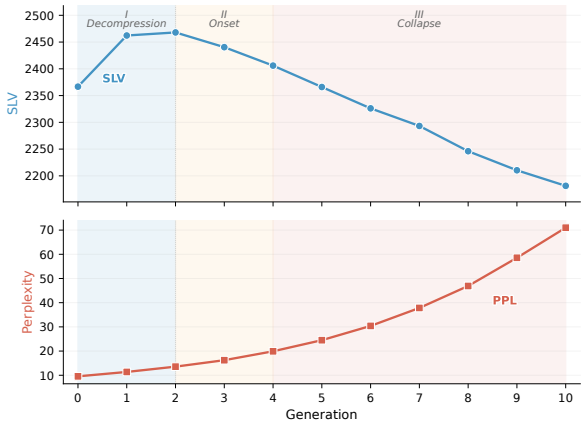


Figure 3: SLV (top) and perplexity (bottom) across 10 recursive generations (r=0, 3-seed mean). In Phase I (Gen 0–2), both metrics rise simultaneously—SLV increase reflects distributional broadening, not model health. From Phase III onward (Gen 4–10), the two metrics diverge: SLV declines monotonically while perplexity accelerates, confirming that post-decompression SLV faithfully tracks structural collapse.

sharper than r_{eff}'s (−1.1% vs. −0.5% in the first post-peak generation), reflecting its sensitivity to distributed contraction across all spectral directions rather than the dominant ones alone. At 1B scale, this sensitivity gap widens to a timing difference: SLV peaks one generation earlier than r_{eff} (see Cross-Scale Pilot below).

Phase III: Accelerating Collapse (Gen 4–10). SLV contraction accelerates: the per-generation decline averages −31.0 during onset (Gen 2–4) vs. −37.4 during acceleration (Gen 4–10), with 78% of the total peak-to-final decline occurring in Phase III (Table 5). By Gen 10, r_{eff} falls to 438.1 (−11.7% from peak), SLV to 2181.4 (−11.6% from peak), and perplexity reaches 71.0 (+643%). Concurrently, σ_1 traces a U-shaped trajectory: after dropping from 1942.8 to 718.7 during decompression, it rises to 874.6 by Gen 10 (Figure 1). This simultaneous σ_1 recovery and r_{eff} decline is a geometric signature of mode collapse—representations concentrate along fewer dominant directions while the overall occupied volume shrinks.

Cross-Scale Pilot. Repeating the protocol on Pythia-1B-deduped (Gen 0–3) reproduces the decompression-onset structure: SLV rises from 2835 to 2906 at Gen 1 (+2.5%) before declining to 2865 at Gen 3, while r_{eff} peaks one generation later at Gen 2 (580.9 vs. 570.8 at Gen 1). This one-generation SLV lead over r_{eff}—absent at 410M where both peak simultaneously—suggests that

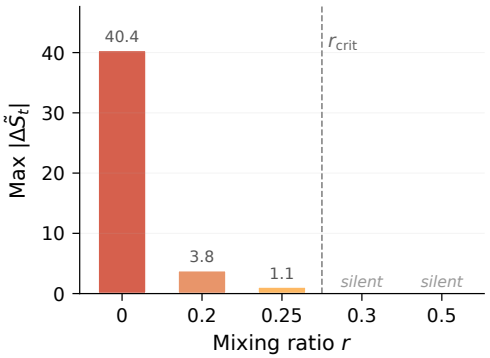


Figure 4: Maximum controller signal magnitude ($\max|\Delta\tilde{S}_t|$, $t \geq 2$) across mixing ratios. The signal drops $10\times$ from r=0 to r=0.2 and is exactly zero for $r \geq 0.3$ ("silent"), rendering rate-based controllers ineffective.

Indicator	Direction at r = 0	Rel Δ%	Reverses under mixing?
SLV	Peak Gen 2, then ↓	11.8%	✓
RankMe	Peak Gen 2, then ↓	4.7%	✓
Token entropy	↓ monotonic	9.6%	✓
Stable rank	↓ monotonic	14.0%	✓
Spectral norm	↑ monotonic	8.0%	✓
Isotropy	↓ noisy	75%	✓
Raw PPL	↑ monotonic	643%	✗
PPL growth	↑ monotonic	—	✓

7/8 indicators reverse under mixing; Raw PPL is the sole exception.

Figure 5: Visual summary of eight EWS indicators under pure-synthetic training (r=0). Direction: trajectory shape at r=0. Rel Δ%: total relative change magnitude. Reverses?: whether the indicator's direction inverts under r=0.5 mixing. Seven of eight indicators reverse; raw PPL is the sole exception.

SLV sensitivity increases with model scale. Perplexity rises monotonically throughout (8.1→13.4, +67%), confirming the downstream-spectral asymmetry at a second scale. Full 10-generation characterization at 1B remains future work.

3.3 Adaptive Mixing Comparison

We evaluate the controller retrospectively by computing its signal on pre-existing fixed-ratio trajectories rather than deploying it in a closed loop; the goal is to characterize signal attenuation, not to propose operational deployment. Under pure synthetic training, SLV peaks at Gen 2 (2468) and declines monotonically; with 50% mixing, SLV rises to 2572 at Gen 10, with only a negligible Gen 1 dip ($\Delta_1 \approx 0.0005$, well below τ). The adaptive controller (k=10) remains silent through all ten generations. Figure 4 quantifies the signal attenuation: the maximum controller signal magnitude drops from 40.4 at r=0 to 3.8 at r=0.2 and reaches exactly zero at $r \geq 0.3$, confirming that even moderate mixing suffices to suppress the rate-based signal (Scheffer et al., 2009).

Despite sustained spectral expansion at r=0.5, downstream metrics degrade: perplexity increases 58% and distinct-4 falls to .152 (below even pure collapse), reinforcing that spectral health does not guarantee model health.

3.4 Alternative EWS Comparison

We compare SLV against seven alternative collapse indicators from three signal sources: embedding geometry (RankMe, isotropy), attention weights (stable rank, spectral norm), and output distributions (token entropy, raw perplexity, perplexity growth rate).

Figure 5 summarizes the comparison under pure-synthetic training (r=0); full tabular data appear in Appendix Table 4. Only SLV and RankMe (Garrido et al., 2023) exhibit a decompression peak at Gen 2 followed by monotonic decline; all other indicators change monotonically from Gen 0 and cannot distinguish collapse onset from the initial fine-tuning transient. SLV's signal-to-noise ratio is 2.5× that of RankMe (11.8% vs. 4.7% relative change).

Seven of eight indicators reverse direction under mixing (Figure 5, last column); raw perplexity is the sole exception. This near-universality confirms that signal attenuation is not specific to SLV: any rate-based controller will fall silent when mixing exceeds r_{crit}. SLV is the only indicator that combines decompression-phase sensitivity, post-peak monotonicity, and sub-second computation cost; it shares a metric foundation with SIGMA's $\log\det G$ ($= 2 \times$ SLV; §4).

3.5 Collapse Phase Diagram

We now map SLV trajectories across mixing ratios to construct the collapse phase diagram defined in §2.3. Figure 6 plots SLV across 10 recursive generations for five mixing ratios; three qualitatively distinct trajectory types emerge:

Type I: Pure Collapse (r=0). SLV exhibits large decompression (+96 points at Gen 1, peaking at Gen 2) followed by accelerating contraction (§3.2).

Type II: Near-critical Mixing (r=0.2, r=0.25). At r=0.2, SLV dips −7.3 points at Gen 2 before recovering to 2467.2 at Gen 10 (+4.3% above

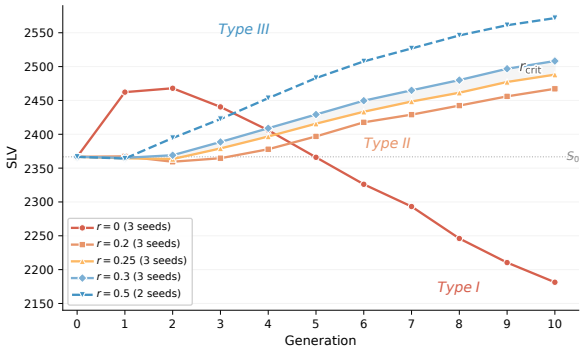


Figure 6: SLV trajectories across mixing ratios (Pythia-410M; r=0, r=0.2, r=0.25, r=0.3: 3-seed mean, solid lines; r=0.5: 2-seed mean, dashed line). Type I (r=0): decompression then monotonic collapse. Type II (r=0.2, r=0.25): transient dip followed by recovery. Type III (r=0.3, r=0.5): monotonic expansion from Gen 2 onward. The shaded band marks the critical threshold $r_{\text{crit}} \in (0.25, 0.3]$. The dashed horizontal line indicates the Gen 0 baseline.

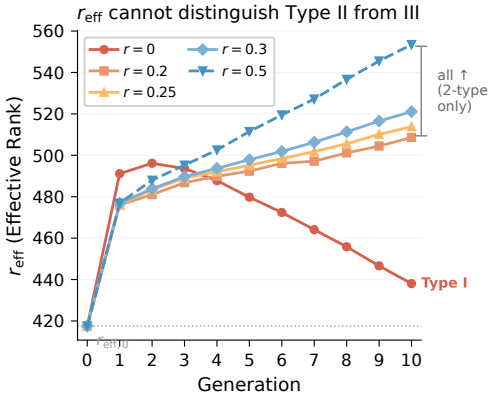


Figure 7: Effective rank (r_{eff}) trajectories across mixing ratios. Unlike SLV (Figure 6), r_{eff} rises monotonically for all $r \geq 0.2$, providing only a 2-type classification and failing to identify the Type II transitional regime.

baseline; full per-generation data in Appendix Table 6). The EMA-smoothed SLV violates non-decreasing at Gen 2, placing r=0.2 below r_{crit}, yet the smoothed signal ($\max \Delta_t = 0.0016$, below τ=0.002) fails to trigger the controller. At r=0.25, the same pattern holds with a weaker dip (-2.1) and $\sim 3\times$ weaker signal. As Figure 7 shows, r_{eff} rises monotonically for all $r \geq 0.2$, yielding only a 2-type classification (collapse vs. non-collapse); SLV's transient dip uniquely identifies the transitional regime.

Type III: Super-critical Mixing ($r \geq 0.3$). SLV rises monotonically from Gen 2 onward. At r=0.3, SLV reaches 2508.2 at Gen 10 (+6.0% above baseline); at r=0.5, 2572 (+8.7%). The

r	Type	SLV	r_{eff}	PPL	D-4	$\max \Delta_t$	Gen.
0	I	2181.4	438.1	71.0	.324	—	10
0.20	II	2467.2	508.1	13.54	.324	.0016	10
0.25	II	2488.1	513.8	13.41	.342	.0005	10
0.30	III	2508.2	521.1	13.34	.295	0	10
0.50	III	2571.6	553.5	15.11	.152	0	10

Table 2: Final-generation spectral and downstream metrics across mixing ratios (all runs to Gen 10). $\max \Delta_t$ computed for $t \geq 2$, excluding the initial fine-tuning transient. Type classification follows §2.3. r=0 reports the mean of three seeds (42, 43, 44); r=0.3 the mean of three seeds (42, 43, 44); r=0.2 the mean of three seeds (42, 43, 44); r=0.5 the mean of two seeds (42, 43); r=0.25 the mean of three seeds (42, 43, 44). D-4 exhibits high cross-seed variance at r=0 (s.d. = 0.15) and r=0.3 (s.d. = 0.155).

critical threshold is robust to EMA smoothing ($\alpha \in \{0.3, 0.5, 0.7\}$ all yield $r_{\text{crit}} \in (0.25, 0.3]$; Appendix Table 7), and the maximum SLV departure below baseline (multi-seed mean) shrinks continuously with r (-7.3 at r=0.2, -3.3 at r=0.25, -1.5 at r=0.3), consistent with a continuous crossover. r_{crit} thus marks a transition in trajectory shape, not in final outcome: Type II trajectories recover spectrally but exhibit an initial contraction transient that Type III trajectories lack.

Output Diversity Divergence. D-4 does not increase monotonically with r: it remains highest around $r \in \{0.2, 0.25\}$ (mean .324–.342, with high cross-seed variance s.d. ≈ 0.17) before dropping to .152 at r=0.5, indicating that spectral expansion does not guarantee diversity. D-4 also shows substantially higher cross-seed variance than SLV (CV = 3.9% vs. 0.12% at r=0.25), confirming that the two metrics capture orthogonal aspects of model behavior.

Table 2 consolidates the final-generation metrics across all five mixing ratios.

Practical Implications. Neither second-order acceleration (Δ_t^2, Eq. 2) nor level-based triggers resolve the attenuation problem: Δ_t^2 is negative in late generations regardless of r, and level-based triggers fail under Type III. The phase diagram provides a direct prescription: setting $r > 0.3$ prevents spectral collapse without per-generation monitoring.

3.6 Cross-Architecture Validation

To test whether the three-phase trajectory generalizes beyond Pythia, we repeat the recursive pipeline

r	Type	SLV_0	SLV_{end}	ΔSLV	Pythia?
0	I	689	415 (G9)	-39.8%	✓
1.0	Ctrl	689	813 (G3)	$+17.9\%$	✓

Table 3: Cross-architecture validation on GPT-2-medium (355M, seed 42). **Pythia?**: whether the trajectory type matches the corresponding Pythia-410M classification. Mixing experiments ($r \in \{0.25, 0.3\}$) are in progress.

on GPT-2-medium (Radford et al., 2019) (355M parameters, different pretraining corpus and tokenizer) under pure synthetic training (r=0, 9 generations) and a real-data control (r=1.0, 3 generations).

Under pure synthetic training, GPT-2-medium reproduces the three-phase pattern (Table 3): SLV peaks at Gen 2 (924, $+34.1\%$ from baseline) and declines to 415 by Gen 9 (-55.1% from peak), while perplexity rises from 15.5 to 329 ($+2028\%$). The σ_1 trajectory follows the same U-shape (6598$\rightarrow$1191 during decompression, rising to 1593 by Gen 9), confirming that dominant-direction reconcentration under falling effective rank is architecture-independent. The collapse magnitude is substantially larger than Pythia's (-55.1% vs. -11.6% SLV decline), reflecting GPT-2-medium's more anisotropic initial spectrum (r_{eff}=23 vs. Pythia's 418).

The real-data control (r=1.0) shows monotonic expansion with perplexity *decreasing* (-27.4%), confirming that collapse is specific to recursive synthetic training. Mixing experiments at $r \in \{0.25, 0.3\}$ are in progress and will determine whether $r_{\text{crit}} \in (0.25, 0.3]$ transfers across architectures.

4 Related Work

Model Collapse. Shumailov et al. (2024) showed that recursive training on model-generated data progressively erodes the tails of learned distributions, a phenomenon they termed *model collapse*. Scaling laws predict tail erosion as a function of recursive depth (Dohmatob et al., 2024b,a; Seddik et al., 2024); the self-consuming loop accelerates collapse even under data accumulation (Alemohammad et al., 2024), with as little as 1% synthetic contamination sufficing (Dohmatob et al., 2025), though preserving all prior real data can delay it (Gerstgrasser et al., 2024); alternative strategies include self-verification filtering of synthetic outputs (Fu et al., 2025) and leveraging weak non-synthetic data (Amin et al., 2025). We complement these output/benchmark-based diagnoses by tracking collapse through the spectral geometry of internal representations, requiring no text generation. Unlike neural collapse (Papyan et al., 2020), which describes last-layer features converging to a simplex ETF in classification, spectral collapse in recursive training reflects progressive embedding volume loss unrelated to class structure.

Spectral Analysis of Representations. Random matrix theory reveals implicit regularization structure in neural network eigenspectra (Martin and Mahoney, 2021); effective rank (Roy and Vetterli, 2007) and RankMe (Garrido et al., 2023) quantify representational dimensionality. The spectral decompression phase, in which singular value spread decreases while effective rank increases, resembles the effect of isotropy regularization (Ethayarajh, 2019); recursive fine-tuning may initially reduce the anisotropy documented in standard training (Gao et al., 2019) before amplifying it during collapse. Recent work has connected spectral properties to training pathologies: He et al. (2025) linked spectral collapse in hidden representations to plasticity loss in continual learning, and Naderi et al. (2025) identified spectral rank collapse in attention matrices as a failure mode of deep transformers. Most directly relevant, Gu et al. (2026) proposed SIGMA, a spectral monitoring framework that tracks singular-value distributions during recursive training on OPT-125M over three generations. Mathematically, our SLV is equivalent to half of SIGMA's $\log\det G$ (both sum log singular values of the embedding matrix), establishing a shared metric foundation. When $n < d$, both metrics face the same rank-deficiency issue; our ϵ-thresholding (§2.1) makes the equivalence approximate rather than exact. Our work extends SIGMA in *scale* (Pythia-410M/1B, 3–8$\times$ larger), *temporal depth* (10 generations vs. 3, enabling the discovery of the three-phase trajectory, signal attenuation, and the collapse phase diagram—all invisible at SIGMA's horizon), and *methodology* (closed-loop SLV-based mixing control; §2.2). At three generations spectral indicators are still rising (Figure 1), which could lead to the premature conclusion that spectral health implies model health. Finally, *signal attenuation*, whereby successful mixing suppresses spectral indicators that would have warned of collapse, is a phenomenon that short-horizon monitoring does not encounter. That SIGMA observed

analogous spectral shifts on OPT-125M, combined with our GPT-2-medium replication (§3.6), provides cross-architecture evidence spanning three model families for the generality of spectral collapse dynamics.

Early Warning Signals and Critical Transitions. Early warning signals (EWS) for critical transitions originate in ecology, where rising autocorrelation and variance precede regime shifts (Scheffer et al., 2009, 2012), with detection methods ranging from statistical tests (Dakos et al., 2012) to deep learning (Bury et al., 2021). A central tension is that stabilization interventions suppress the very precursors they rely on (Scheffer et al., 2009, 2012). Rate-induced critical transitions (Siteur et al., 2016) further demonstrate that interventions can themselves alter system dynamics. We encounter a parallel tension: real-data mixing reverses the spectral trajectory entirely (§2.2), unlike ecological stabilization which preserves signal direction; our contribution is quantifying how rapidly this signal degrades across mixing ratios. He et al. (2025) provides a bridge between these domains by connecting spectral collapse to critical transitions in continual learning.

Data Mixing for LLM Training. Data composition significantly affects LM quality (Longpre et al., 2024; Muennighoff et al., 2023), and several methods optimize domain mixtures through proxy models or predictive formulas (Xie et al., 2023; Ye et al., 2025; Chen et al., 2025). These approaches evaluate mixing through loss or benchmarks. Our work reveals a complementary dimension: real-vs-synthetic mixing reshapes the spectral geometry of internal representations, producing a phase diagram (§3.5) that loss-based metrics do not capture, and our controller targets recursive spectral collapse rather than general domain balance.

5 Discussion and Conclusion

Decompression Mechanism. The spectral decompression phase reflects fine-tuning's redistribution of variance across embedding dimensions. Pretrained Pythia-410M exhibits strong anisotropy (σ_1=1943, dominating r_{eff}=418 of 1,000 possible directions); early recursive fine-tuning compresses this dominant direction (σ_1 drops 63% by Gen 2) while inflating subordinate directions, increasing both r_{eff} and SLV. This resembles the isotropy increase observed during standard fine-tuning (Ethayarajh, 2019; Martin and Mahoney, 2021), reversed here by recursive amplification. That downstream performance degrades during this nominally beneficial spectral expansion confirms that these indicators are collapse detectors, not quality measures—their value lies in post-peak monotonicity.

Prevention–Detectability Tradeoff. In ecology, stabilization interventions suppress the rising variance that serves as an early warning signal (Scheffer et al., 2009, 2012); our spectral setting exhibits a stronger form—real-data mixing reverses the trajectory entirely, causing SLV to rise rather than fall. Seven of eight EWS indicators exhibit this reversal (§3.4), confirming that the tradeoff is structural rather than metric-specific. The controller signal drops $10\times$ from r=0 to r=0.2 and vanishes at $r \geq 0.3$ (Figure 4), leaving no regime where both prevention and detection coexist.

Practical Deployment. Although reactive monitoring fails under signal attenuation, the phase diagram provides a direct alternative: calibrating r_{crit} via a pilot pipeline (10 generations at 3–5 mixing ratios, ∼50–80 single-GPU hours for a 410M model) and then setting $r > r_{\text{crit}}$ eliminates per-generation monitoring. Spectral stability does not guarantee downstream quality—perplexity still increases 58% at r=0.5—but degradation under Type III is gradual and predictable, unlike the accelerating collapse below r_{crit}.

Conclusion. Spectral monitoring of recursive synthetic training reveals a reproducible three-phase collapse trajectory, identifies SLV as a computationally cheap collapse indicator, and exposes signal attenuation that limits reactive controllers. A collapse phase diagram localizes $r_{\text{crit}} \in (0.25, 0.3]$ for Pythia-410M, with preliminary cross-architecture support from GPT-2-medium (§3.6) and Pythia-1B. Setting $r > r_{\text{crit}}$ prevents spectral collapse without per-generation monitoring, reducing the problem to one-time offline calibration.

6 Limitations

Scale and Replication. Multi-seed replication covers pure collapse (three seeds), r=0.25 (three seeds), and r=0.3 (three seeds); r=0.2 reports the mean of three seeds (42, 43, 44); r=0.5 reports the mean of two seeds (42, 43). SLV is computed from n=1,000 held-out embeddings; bootstrap resampling yields a coefficient of variation below 0.2%

at this sample size, and CV remains below 0.4% across $n \in \{250, 5000\}$. We validate at two model scales only (Pythia-410M and a 1B pilot through Gen 3); while SIGMA observed analogous spectral shifts on OPT-125M (Gu et al., 2026), systematic cross-architecture validation remains future work. The 1B pilot covers only three generations, which is sufficient to confirm the decompression peak but insufficient to characterize the full acceleration phase or to estimate r_{crit} at that scale.

Domain and Horizon. All fine-tuning uses a single data domain (OpenWebText, English web text). Spectral collapse dynamics may differ substantially for code, multilingual, or domain-specific corpora, where the structure of the data manifold and the degree of distributional shift between real and synthetic text vary. Ten recursive generations may be shorter than production pipelines, and the spectral trajectory beyond this horizon is unknown. It is possible that trajectories currently classified as Type II (transitional) eventually revert to collapse at longer horizons, which would shift the effective r_{crit} upward. Signal attenuation may extend to implicit-loop settings where web corpora contain model-generated content, but validating this generalization requires controlled experiments with web-scale data mixtures.

Embedding Layer. All spectral metrics are computed from last-layer embeddings; intermediate layers may exhibit different collapse dynamics, and exploring layer-wise spectral trajectories is future work.

Controller Design. The adaptive controller uses a simple proportional design with a single gain parameter k and a fixed threshold τ, without integral or derivative terms. A more sophisticated controller incorporating cumulative drift or rate-of-change signals might partially mitigate signal attenuation, but exploring such designs is beyond the scope of this work.

Training and Compute Details. Fine-tuning uses AdamW (lr 5×10^{-5}, weight decay 0.01, linear warmup 5%, effective batch size 32) for one epoch per generation, with gradient clipping at max norm 1.0. Synthetic text is generated with temperature 1.0, top-k=50, max length 512. All experiments run on NVIDIA RTX PRO 6000 GPUs (96 GB); each 10-generation pipeline requires 15–26 single-GPU hours depending on the mixing ratio, totaling approximately 220 GPU-hours across all reported experiments.

Code and Data. Code and data will be released upon acceptance.

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A Alternative EWS Indicator Details

Table 4 provides full tabular data for the eight alternative early warning indicators compared in §3.4.

Indicator	Source	Peak	Mono?	Rel Δ	Rev.?
SLV	emb-SVD	Gen 2	✓	11.8%	✓
RankMe	emb-SVD	Gen 2	✓	4.7%	✓
Token ent.	logit	—	✓	9.6%	✓
Stable rank	attn wt.	—	✓	14.0%	✓
Spec. norm	attn wt.	—	✓	8.0%	✓
Isotropy	emb	—	× (6)	75%	✓
Raw PPL	eval	—	✓	704%	×
PPL growth	ppl	—	∼ (1)	—	✓

Table 4: Alternative early warning indicators under pure-synthetic training (r=0, seed 42, Pythia-410M, 10 gen). **Peak**: generation at which the indicator reverses (— = monotonic from Gen 0). **Mono?**: post-peak monotonicity (✓ = yes; number = violations). **Rel Δ**: magnitude of total relative change; for peaked indicators (SLV, RankMe), computed as $|v_{10} - v_{\text{peak}}|/|v_{\text{peak}}|$; for monotonic indicators, $|v_{10} - v_0|/|v_0|$. **Rev.?**: direction reversal under r=0.5 mixing (✓ = reversed).

B Per-Generation Cross-Seed Variance

Table 5 reports all six metrics at every generation under pure-synthetic training (r=0), with mean and sample standard deviation across three seeds (42, 43, 44). Spectral metrics (r_{eff}, SLV, σ_1) exhibit low cross-seed variance throughout the trajectory (SLV coefficient of variation $< 1.3\%$ at all generations), confirming that spectral collapse follows a deterministic geometric trajectory. In contrast, D-4 variance increases sharply after Gen 5, reaching a coefficient of variation of 46% at Gen 10, consistent with the stochastic nature of n-gram diversity under progressive mode collapse.

Table 6 provides the same breakdown under transitional mixing (r=0.2), where D-4 variance is the dominant source of cross-seed disagreement.

C Controller Signal Sensitivity to EMA Smoothing Parameter

D Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform the related-work section.

Table 5: Per-generation metrics under pure-synthetic training (r=0, Pythia-410M, mean$_{\pm \text{s.d.}}$ across seeds 42, 43, 44). Gen 0 shares a single pretrained checkpoint; D-4 variance at Gen 0 reflects generation sampling. Bold marks the spectral peak.

Gen	r_{eff}	SLV	PPL	HeSw	D-4	σ_1
0	$417.5_{\pm 0.0}$	$2366.7_{\pm 0.0}$	$9.55_{\pm 0.00}$	$.413_{\pm .000}$	$.738_{\pm .006}$	$1942.8_{\pm 0.0}$
1	$491.1_{\pm 0.6}$	$2462.3_{\pm 2.7}$	$11.37_{\pm 0.01}$	$.390_{\pm .001}$	$.609_{\pm .014}$	$808.9_{\pm 5.4}$
2	$\mathbf{496.2}_{\pm 0.3}$	$\mathbf{2468.0}_{\pm 1.7}$	$13.56_{\pm 0.02}$	$.380_{\pm .004}$	$.505_{\pm .011}$	$718.7_{\pm 7.0}$
3	$493.6_{\pm 0.4}$	$2440.4_{\pm 2.0}$	$16.23_{\pm 0.02}$	$.367_{\pm .005}$	$.465_{\pm .033}$	$708.7_{\pm 7.5}$
4	$487.8_{\pm 1.9}$	$2406.0_{\pm 2.0}$	$19.90_{\pm 0.09}$	$.354_{\pm .003}$	$.419_{\pm .047}$	$722.9_{\pm 6.9}$
5	$479.8_{\pm 0.5}$	$2366.0_{\pm 3.8}$	$24.50_{\pm 0.17}$	$.341_{\pm .003}$	$.412_{\pm .053}$	$749.2_{\pm 7.1}$
6	$472.4_{\pm 0.6}$	$2326.1_{\pm 8.6}$	$30.40_{\pm 0.45}$	$.329_{\pm .004}$	$.402_{\pm .067}$	$764.6_{\pm 5.7}$
7	$464.1_{\pm 0.2}$	$2293.2_{\pm 10.2}$	$37.82_{\pm 0.94}$	$.319_{\pm .000}$	$.375_{\pm .082}$	$788.4_{\pm 4.0}$
8	$455.8_{\pm 1.1}$	$2246.1_{\pm 7.4}$	$46.90_{\pm 2.41}$	$.311_{\pm .005}$	$.337_{\pm .137}$	$806.4_{\pm 8.0}$
9	$446.6_{\pm 2.1}$	$2210.4_{\pm 23.1}$	$58.56_{\pm 4.52}$	$.301_{\pm .003}$	$.316_{\pm .148}$	$841.1_{\pm 23.6}$
10	$438.1_{\pm 4.8}$	$2181.4_{\pm 28.3}$	$70.99_{\pm 5.56}$	$.296_{\pm .006}$	$.324_{\pm .150}$	$874.6_{\pm 44.2}$

Table 6: Per-generation metrics under transitional mixing (r=0.2, Pythia-410M, mean$_{\pm \text{s.d.}}$ across seeds 42, 43, 44). Gen 0 shares a single pretrained checkpoint; D-4 variance at Gen 0 reflects generation sampling. HellaSwag was not evaluated for mixing experiments.

Gen	r_{eff}	SLV	PPL	D-4	σ_1
0	$417.5_{\pm 0.0}$	$2366.7_{\pm 0.0}$	$9.55_{\pm 0.00}$	$.738_{\pm .006}$	$1942.8_{\pm 0.0}$
1	$476.2_{\pm 0.8}$	$2367.4_{\pm 2.1}$	$10.28_{\pm 0.00}$	$.671_{\pm .015}$	$961.8_{\pm 1.1}$
2	$481.7_{\pm 1.4}$	$2359.4_{\pm 1.5}$	$10.74_{\pm 0.01}$	$.519_{\pm .037}$	$879.2_{\pm 10.3}$
3	$486.7_{\pm 0.3}$	$2364.6_{\pm 1.8}$	$11.14_{\pm 0.01}$	$.471_{\pm .006}$	$811.9_{\pm 9.6}$
4	$489.6_{\pm 0.6}$	$2377.9_{\pm 1.2}$	$11.49_{\pm 0.00}$	$.429_{\pm .026}$	$779.2_{\pm 9.2}$
5	$492.6_{\pm 0.5}$	$2396.8_{\pm 1.8}$	$11.83_{\pm 0.00}$	$.420_{\pm .037}$	$748.8_{\pm 9.3}$
6	$495.5_{\pm 1.2}$	$2417.7_{\pm 10.5}$	$12.17_{\pm 0.02}$	$.413_{\pm .070}$	$749.7_{\pm 22.5}$
7	$496.7_{\pm 1.2}$	$2429.0_{\pm 6.4}$	$12.49_{\pm 0.05}$	$.379_{\pm .096}$	$750.9_{\pm 30.6}$
8	$499.8_{\pm 2.3}$	$2442.4_{\pm 10.3}$	$12.82_{\pm 0.12}$	$.365_{\pm .119}$	$753.9_{\pm 23.4}$
9	$503.8_{\pm 1.2}$	$2456.0_{\pm 6.3}$	$13.18_{\pm 0.18}$	$.349_{\pm .143}$	$738.1_{\pm 37.6}$
10	$508.1_{\pm 2.8}$	$2467.2_{\pm 12.0}$	$13.54_{\pm 0.25}$	$.324_{\pm .171}$	$722.7_{\pm 26.6}$

α	r=0	r=0.2	r=0.25	r=0.3	r=0.5
0.3	.0141	.0009	.0003	0	0
0.5	.0171	.0016	.0005	0	0
0.7	.0183	.0023	.0005	0	0

Table 7: Normalized maximum controller signal magnitude ($\max |\Delta \tilde{S}_t| / \tilde{S}_t$, $t \geq 2$) across mixing ratios and EMA smoothing parameters. The critical threshold $r_{\text{crit}} \in (0.25, 0.3]$ is invariant to α: signal is exactly zero for $r \geq 0.3$ at all three smoothing values, confirming that the phase boundary is a structural property of the trajectory rather than an artifact of smoothing.